

Ordinary Portland Cement · Specification (Sample)

1. SCOPE

This standard prescribes the requirements and the related methods of sampling and testing for ordinary Portland cement used in general concrete construction. This standard applies to 33, 43 and 53 grade ordinary Portland cement.

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2. REFERENCES

The following standards are referred to in this standard: IS 4031 for methods of physical tests on hydraulic cement, and IS 4032 for methods of chemical analysis of hydraulic cement.

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3. MATERIALS AND MANUFACTURE

Cement shall be manufactured by intimately mixing together calcareous and argillaceous materials, burning them at clinkering temperature, grinding the resultant clinker, and adding a small percentage of gypsum at the grinding stage.

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4. REQUIREMENTS

4.1 Fineness · The residue on 90-micron sieve shall not exceed 10 percent when tested per IS 4031 Part 1. 4.2 Setting time · The initial setting time shall not be less than 30 minutes and the final setting time shall not exceed 600 minutes when tested per IS 4031 Part 4. 4.3 Soundness · The expansion of cement paste shall not exceed 10 mm when tested by Le Chatelier method per IS 4031 Part 3. 4.4 Compressive strength · The compressive strength of 53 grade cement shall be not less than 53 MPa at 28 days when tested per IS 4031 Part 6.

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5. CERTIFICATION AND MARKING

Every bag of cement shall be legibly and indelibly marked with the manufacturer's name, the grade of cement, the week and year of manufacture, and the BIS certification mark (ISI mark) under a valid BIS licence per the BIS Act 2016. Cement conforming to this standard may be marked with the Standard Mark under a BIS licence; the manufacturer must obtain a licence from BIS before applying the mark.

5.1 Licence application · The manufacturer shall apply to BIS in the prescribed form with details of manufacturing process, quality control setup and test facilities.

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6. SAMPLING AND TESTS

The purchaser may sample cement per IS 3535. All tests shall be carried out in accordance with the methods given in IS 4031.